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A machine learning-based approach supported by HJ-biplot statistical analysis reveals novel avenues of artificial intelligence in oncological research

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ABSTRACT

Artificial intelligence (AI) is being used in oncology at an exponential rate. However, unless the intricacies of its research landscape are clarified, AI's potential in cancer research may remain unrealized. To circumvent the bias associated with human-assisted review, we adopted a combination of Latent Dirichlet Allocation (LDA) with HJ-Biplot to examine 11149 documents published in Web of Science from 2003 to 2023. The probabilistic topic LDA algorithm automatically found up to 32 research thematic trends with the highest semantic quality. The HJ-Biplot further confirmed that automated imaging and computer-assisted diagnosis of tumors were emerging scientific fields showing great promise in guiding precision oncology. Furthermore, the analysis highlighted the AI's capacity to assess a wide range of data types, which is necessary for cohort studies to develop new predictive models. Other AI applications included evaluating treatment response, testing novel molecular therapies in clinical trials, laparoscopy, and robotics in surgery. The socioeconomic and cultural impact of cancer were neglected topics that require additional AI research. Consequently, we proposed an enhanced topic analytical framework more accurate than simple probabilistic topic modeling and bibliometrics in examining novel research paths, because the LDA model gained an extra analytical boost using the HJ-Biplot tool.

1. Introduction

Cancer remains a formidable challenge to global health, with projections estimating 30.2 million new diagnoses annually by 2040 [1,2]. Oncology is the branch dedicated to the study of the intricate details of cancer including pathology, laboratory medicine, radiology, and reporting on molecular diagnostics. Although unprocessed clinical data can yield insightful information from cancer, when it comes to combining the heterogeneous multi-modal decode data that comes from the extensive research on cancer, artificial intelligence (AI) models can handle large amounts of data to identify patterns and correlations undetectable to human clinicians [3–6].

AI technology, especially its subfields in machine learning (ML) and deep learning (DL), has significant advantages for oncology. Given the volume of scientific literature on cancer, AI expedites the extraction of vital information. Precision oncology, which makes use of tumor biomarker data to guide early diagnosis and tailored treatment, is supported by AI. Varied ML/DL applications can enhance and advance cancer imaging. Based on population-level cancer monitoring data, AI can even deconstruct complex epidemiological, socio-economic and cultural factors on cancer incidence. AI promises a thorough assessment catered to each person's clinical profile [7] by better integrating various diverse data streams from cancer patients. Lastly, AI aids in finding new oncological targets, accelerating drug discovery, repurposing existing

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therapies, and even formulating new hypotheses [8]. To reduce bias and guarantee reproducibility of AI algorithms in oncology, some critical challenges must be addressed including insufficient clinical validation and a lack of standardized research reporting among others [6]. Unfortunately, the entire spectrum of AI-based oncological research remains unmapped.

Investigations on scientific mappings, bibliometric methods, and network visualization studies [9] have revealed a research landscape of the relationships between oncology knowledge and AI [10–18]. However, traditional bibliometric analyses often fall short of fully quantifying research quality because its metrics rely mostly on citation-tracking databases and do not permit comprehensive coverage of research publications [9]. Recently, large language models (LLMs) have been leveraged to map this scientific landscape with greater accuracy than traditional bibliometrics [19]. Latent Dirichlet Allocation or LDA is a type of unsupervised machine learning (ML) algorithm that can automatically discover hidden or latent thematic structures (i.e., topics) from large collections of documents [20]. LDA's unsupervised nature allows documents to emerge organically from the text and for topics to emerge without human intervention based on their statistical properties [21]. Combining LDA [20] with the statistical technique HJ-Biplot [22] may give an extra analytical boost that may greatly accelerate our request for study patterns in oncological investigations using AI.

We propose an enhanced topic analytical framework as an alternative to classic bibliometrics to present a comprehensive and up-to-date summary of the research trends and knowledge gaps in AI applications in Oncology. We adopted this integrated approach to answer the following questions: What are the primary research topics in Oncology (Q1), how have oncology research questions evolved over time (Q2), and how are these research topics distributed across countries and scientific journals (Q3)?

2. Methods

2.1. Data collection and search strategy

The study dataset was retrieved from Web of Science Core Collection (specifically the Science Citation Index Expanded, or SCIE), powered by Clarivate. The study inclusion criteria were all the research documents, written in English in peer-reviewed journals, that were published between 2003 and 2023 and that dealt with the application of AI in oncology. According to the inclusion criteria, the earliest article we found in our search was published in 2003. To ensure dataset cleanliness/homogeneity, books, book chapters, grey literature, and reports were excluded. The large sample size offset any potential biases in the search approach. Previous studies [11,13–17] served as the foundation for the search method. The search was conducted on April 13, 2023. The search string, which used a Boolean computation, was as follows: (((TI=((tumor* or tumour* or cancer* or carcin* or oncolog* or neoplas* or malignan* or carcinoma* or adenocarcinoma* or choricarcinoma* or leukemia* or leukemia* or metastat* or sarcoma* or teratoma* or melanoma* or lymphoma* or myeloma* or oncogene* or chemotherap*))) and TI=((“Artificial intelligence” or “Machine intelligence” or “artificial neutral network*” or “Machine learning” or “Deep learn*” or “Natural language process*” or “Robotic*” OR “thinking computer system” or “fuzzy expert system*” or “evolutionary computation” or “hybrid intelligent system*”))) and LA=(English)) and DT=(Article or Review). Our search procedure was in line with the 'Preferred Reporting Items for Systematic Reviews and Meta-Analyses' (PRISMA) 2020 guidelines [23]. The lead author, in collaboration with the co-authors, reviewed the full metadata of all potentially eligible records exported from the Web of Science Core Collection. The team evaluated each article's relevance and bibliometric contributions based on the title, abstract, publication year, journal, author, and institutional affiliation. The articles that meet the predefined inclusion criteria were compiled into a dataset and stored as *TXT* files that contained a total of 11,149 different articles.

The annual growth rate of publications was calculated using the Bibliometrix package in R, which computes the Compound Annual Growth Rate (CAGR) using the following formula:

$$CAGR = \left(\frac{V_f}{V_i}\right)^{\frac{1}{n}} - 1 \quad \text{(Eq1)}$$

where V_f: was the number of publications in the final year of the study period, V_i: was the number of publications in the initial year of the study period, and n was the number of years between the initial and final year.

2.2. Latent Dirichlet Allocation (LDA)

For topic modelling in this study, we utilized Latent Dirichlet Allocation (LDA) as our initial method. As a generative Bayesian probabilistic model, LDA assumes that documents contain a variety of themes of differing degrees. This method is known as a mixed-membership model in statistical analysis and enables the identification of the wide range of research topics seen in scientific publications [24]. Topics in LDA are thought of as multinomial distributions of vocabulary terms, in which each word has a given probability of occurring in a topic. Words are weighted by their conditional probabilities, creating clusters that reflect specific underlying themes or issues. Importantly, prior knowledge of the topics or how they are distributed across the documents is not necessary for LDA to function. Instead, these elements simply derived from the statistical properties of the data and the foundational principles of the model.

The LDA model was validated using the topic coherence metric, which evaluates the semantic interpretability of generated topics from a human perspective [25]. This validation was achieved by analyzing the frequency and co-occurrence of keywords within each topic. We utilized the R package textmineR for text mining and topic modeling, which facilitated the optimization of the ideal number of topics (k). Since higher scores indicate better interpretability, we selected the model with the highest coherence score within the analyzed range (k = 4-50), ensuring a balance between granularity and thematic clarity.

Topic coherence was evaluated using the Cv metric, which quantifies the semantic similarity of top-ranked words within a given topic. This metric calculates coherence based on word co-occurrence statistics in a reference corpus, aligning computational outputs with human semantic interpretation. Specifically, Cv was computed by examining pairs of high-probability words within each topic, with higher scores indicating better interpretability. The calculation of coherence scores were streamlined using the textmineR package. A model with the highest coherence score among those in the study rank (from k = 4-50) was chosen to achieve a balance between granularity and thematic clarity since higher scores indicate better interpretability.

2.3. Identifiying reaserch topics

The LDA topic identification methodology was structured into three distinct stages: (i) data preprocessing, (ii) construction of the LDA model, and (iii) topic annotation. For the first two stages, we utilized LDAShiny [26], a robust open-source R package that facilitates conducting scientific literature reviews through a web-based interface. This interface integrates machine learning algorithms to enhance the analytical process and implements LDA modelling via Bayesian inference.

2.3.1. Preprocessing

Bigrams, or sequential word pairs, were utilized in the preprocessing stage to tokenize every abstract. During the normalization workflow, the text was converted to lowercase, and any punctuation, hyphens, brackets, numerals, whitespace, and other non-alphanumeric characters were removed. Furthermore, a predetermined list of stop words – such

as articles and prepositions – was filtered out as they primary support grammatical structure rather than convey semantic meaning. The list of stop words was obtained from standard libraries such as NLTK (i.e., natural language tool kit) and Snowball and was adapted to include irrelevant terms specific to the medical and technical fields.

2.3.2. Construction of the LDA model

Topic models use the relationships between words and the underlying semantic themes in a corpus of documents as latent variable frameworks for text analysis [27]. Finding the optimal number of topics, or latent variables inside a corpus requires striking a balance between a higher number of topics for thorough coverage and a more focused set for clarity. Using a Gibbs sampling inference method across 500 iterations [20], we ran simulations varying the topic count (k) from 4 to 50 in increments of two to evaluate the performance of the LDA model. The efficacy of the LDA model was assessed primarily through topic coherence [28,29]. This metric measures a model's interpretability from a human-centric perspective and is widely recognized as more revealing than traditional computational metrics like perplexity [30].

2.3.3. Assigning labels to topics

Although the LDA model does not automatically assign semantic labels to topics, computational methods enable the comprehension of nuanced meanings found in human language. Consequently, in topic modeling efforts, human-assisted annotation has become the norm [31]. To ensure accurate semantic delineation of topics, we utilized a manual labeling approach. The process involved analyzing two primary sources of information: lists of the most frequently occurring words with the highest probability within each topic, and a selection of representative titles associated with these topics. This dual-source strategy facilitated a comprehensive understanding of the thematic structure. A diverse group of six experts, including the authors and independent researchers with backgrounds in bibliometric analysis and oncological research, carried out the manual topic labeling. Two sources of data were given to the team: lists of the most commonly occurring words produced by the model and samples of three document titles with their algorithm-classified summaries. They were tasked with independently validating and summarizing the identified topics, as well as reviewing existing literature to uncover research trends, gaps, and key influential studies. Any disagreements among team members were resolved through remote communication to reduce bias, enhance rigor, and ensure that the thematic structures and semantic interpretations aligned with the research objectives. Reference [31] offers a guide to the procedures used to ensure the trustworthiness of the assigned labels.

2.4. Quantitative indices

To validate and summarize the identified topics, we selected the three articles that demonstrated the strongest ties to each topic based on their content alignment and thematic coherence. The selection of these articles per topic was based on their high probability of association with the respective topic, as determined by the LDA model. These articles represented the most relevant and thematically aligned examples within each topic, ensuring clarity and interpretability. Limiting the selection to three articles provided a focused and manageable summary of the key characteristics of each topic without overwhelming the analysis or the reader. This approach struck a balance between thorough representation and practical interpretability. These articles were then condensed into concise summaries, capturing the essence of each topic. This manual approach not only provided a gold-standard reference [31] but also ensured interpretability and relevance in the context of the study.

The main outputs produced by the LDA model are distributions of topics across documents and words across topics. However, the large number of documents and words makes it difficult to intuitively understand underlying topics or emerging trends. We used quantitative indices, as recommended by Xiong et al. [32], to overcome this challenge. These indices improve the interpretability of the results and insights obtained from the analysis by synthesizing the document-topic and topic-word distributions across many layers.

The derived metrics are characterized as follows: the temporal distribution of topics is determined by analyzing their evolution over specified periods.

$$\theta_k^y = \frac{\sum_{m \epsilon y} \theta_{mk}}{n^y} \quad \text{(Eq2)}$$

in which $m\epsilon y$ denotes articles released within a specified year, θ_{mk} signifies the fraction of the k-th topic within each document, and n^y corresponds to the aggregate count of articles disseminated in that year.

The distribution of topics across journals was quantified by the ratio j: θ_k^j representing the prevalence of the k-th topic within journal j.

$$\theta_k^j = \frac{\sum_{m \epsilon j} \theta_{mk}}{n^j} \quad \text{(Eq3)}$$

where, $m\epsilon j$ denotes the articles within a specific journal, θ_{mk} indicates the proportion of the k-th topic in each document, and n^j represents the cumulative number of articles issued in journal j.

The distribution of topics across countries was quantified as the ratio c: θ_k^c, which signifies the presence of the k-th topic within country c.

$$\theta_k^c = \frac{\sum_{m \epsilon c} \theta_{mk}}{n^c} \quad \text{(Eq4)}$$

where, $m\epsilon c$ denotes the articles associated with a specific country, θ_{mk} indicates the proportion of the k-th topic within each document, and n^c represents the total count of articles published within country c.

2.5. Statistics

We conducted a linear regression analysis, designating the year as the independent variable and the annual proportions of topics as the dependent variable to elucidate the trajectory of topic trends [33]. From the regression slopes, we determined the directionality of these trends and set a significance threshold of $p < 0.01$ (Eq (1)). Topics that showed statistically significant positive slopes were identified as having upward trends, while those with statistically significant negative slopes were considered to be in decline.

2.6. HJ-biplot

Like a scatter plot that shows the combined distribution of two variables, biplots are graphical representations of multivariate data that allow visualization of three or more variables. They were first described by Gabriel [34]. Among Gabriel's most renowned biplot factorizations are the GH-Biplot and JK-Biplot. The GH-Biplot is designed to explore relationships among columns (variables) with a focus on superior representation quality. Conversely, the JK-Biplot evaluates the similarity among rows (individuals) with an emphasis on higher representation quality. To improve the analysis of the LDA model, the HJ-Biplot technique [22] has been proposed as an alternative to traditional biplots. Its goal is to simultaneously optimize the representation quality of both rows and columns within the same low-dimensional space [28,29,35].

The HJ-Biplot [22,36], derived from the classical biplots introduced by Gabriel [34], serves as an exploratory tool for detecting patterns within a data matrix and graphically displaying the relationships between rows (years) and columns (topics) [22]. This multivariate technique enables precise evaluation by emphasizing connections among components, years, and topics. The HJ-Biplot analysis was conducted using the Multbiplot software [37], which facilitates the simultaneous visualization of rows (years, journals, or countries) and columns (topics) in a shared low-dimensional space. The input data matrix was structured with rows representing the subjects (e.g., years or journals) and columns

representing the variables (topics or keywords). Singular value decomposition (SVD) was applied to this matrix to project both rows and columns as vectors onto a two-dimensional plane, optimizing representation quality. The probability of occurrence of keywords was calculated based on their weighted frequencies across documents. Each keyword's contribution to a topic was determined by normalizing its frequency within the topic relative to its overall presence across all topics. This normalization ensured that the probabilities reflected the keyword's importance to specific topics, enabling accurate multivariate analysis.

The data visualization involved representing a multivariate data matrix X_{nxp} using two sets of vectors: markers $g_1, g_2, \ldots, g_n$ for rows and markers $h_1, h_2, \ldots, h_p$ for columns. Each row corresponded to a subject, and each column to a variable, allowing both marker sets to be projected onto a shared reference system with optimal representation quality. With matrix A defining markers $g_1, g_2, \ldots, g_n$ and matrix B defining markers $h_1, h_2, \ldots, h_p$, the relationship was expressed as $X = AB^T$.

The markers were derived through the singular value decomposition (SVD) of the data matrix X, expressed as $X = UDV^T$ Here, U contains the eigenvectors of XX^T, V holds the eigenvectors of X^TX, and D is a diagonal matrix of the singular values σ_i of X. Matrices A and B correspond to the first two columns of UD and VD, respectively.

Combining knowledge from various approaches, such as Multidimensional Scaling, Correspondence Analysis, Factor Analysis, and classical Biplots, is necessary to interpret the HJ-Biplot [22]. In particular, the lengths of column markers (vectors) roughly represent the standard deviations, whereas the distances between row markers indicate their similarity. Approximations of correlations are provided by the cosines of the angles formed by column vectors. Acute angles denote a strong positive correlation, obtuse angles a negative correlation, and right angles uncorrelated variables, according to the Multbiplot software [37].

3. Results

3.1. Overview of the dataset

A total of 12,445 documents from 1717 sources between 2003 and 2023 were found in the initial search. After removing 133 duplicates and identifying 1151 documents without available abstracts, along with 12 lacking affiliation information, a final dataset resulting in a total of 11,149 documents was compiled for analysis (Fig. 1). Table 1 offers a thorough summary overview of publications pertaining to oncology and AI, giving readers a comprehensive understanding of the state of research in this field.

An impressive annual growth rate of 39.46% was noted, indicating the increasing interest in this area of study over time. As a gauge of how rapidly the literature in the selection was expanding, the study showed

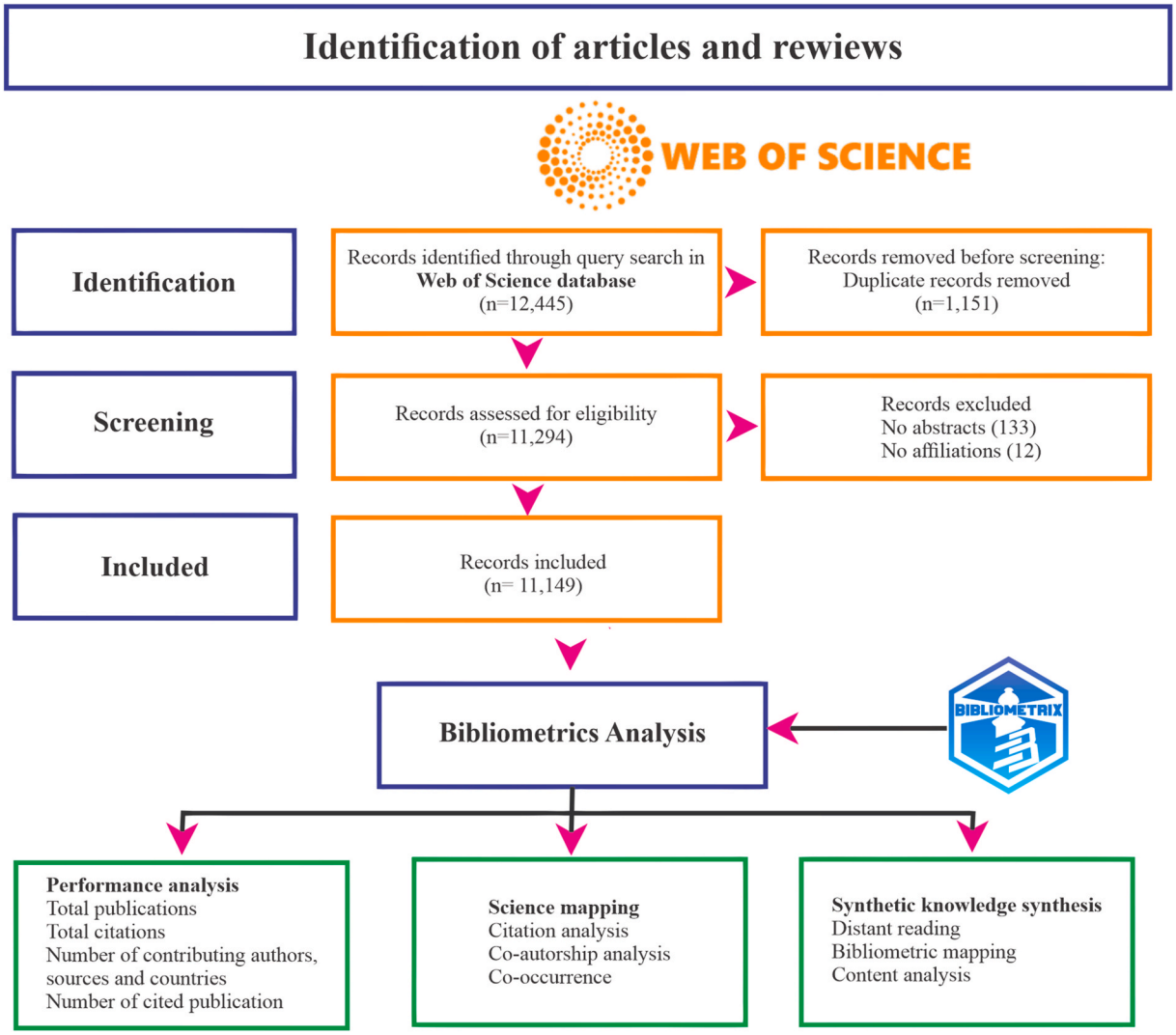


Fig. 1. PRISMA flow diagram illustrating the workflow for article selection and bibliometric analysis in oncology using the Web of Science database.

Table 1
Comprehensive overview of data characteristics and publication metrics on AI in oncology (2003–2023).

Description	Results
Main Information about Data	
Timespan	2003:2023
Sources	1717
Documents	11149
Annual Growth Rate %	39.46
Document Average Age	3.51
Average citations per doc	18.46
References	240322
Document Contents	
Keywords Plus	8635
Author's Keywords	13679
Authors	
Authors	47780
Authors of single-authored docs	139
Authors Collaboration	
Single-authored docs	145
Coauthors per Doc	7.73
International co-authorships (%)[a]	24.37
Document Types	
Article	9755
Review	1394

Retrieved from the Clarivate Analytics' Citation Index Expanded database.

[a] International co-authorship: when at least one co-author was based in a country different from the corresponding author's institution.

that the average age —or the time since the publication —of the examined articles was 3.51 years. The contribution of the articles to the field was further highlighted by an average of 18.46 citations per article.

The research covered a breadth of themes, as evidenced by identification of 13,679 author-provided keywords and 8635 additional keywords related to the content of the documents. Just 139 documents had a single author, out of a total of 47,780 coauthors. With an average of 7.73 coauthors per document and a notable 24.37% of international collaborations (when at least one coauthor was based in a country different from the corresponding author's institution), author collaboration (co-authorship within a publication, regardless of geographic location) was prevalent and demonstrated the global and multidisciplinary nature of research in this field. Regarding document types, the bulk of publications were articles (9755), with reviews following closely behind (1394). This suggests that the focus was on producing both original knowledge as well as synthesizing and reviewing previous research.

The 11,149 documents published in this field featured authors from 92 different countries or regions. The top 30 countries with the highest rankings are shown in Table 2, which highlights their relative contributions to knowledge. Leading the way in this field of research are China and the United States, with 2877 and 2770 publications, respectively. Combined, these countries account for almost 46% of all articles published, with India (6.57%) and South Korea (6.23%) following thereafter. China, South Korea, Japan, India, Taiwan, and other Asian countries are among the top ten contributors to AI research in oncology. Similarly, European nations are well-represented; among the top 15 contributors are Italy, Germany, the UK, and the Netherlands.

Research on AI applications in oncology was published in 1717 scholarly journals overall. With 467 published articles, *Frontiers in Oncology* topped the list, closely followed by *Cancers* (399) and *Scientific Reports* (316) (see Table 3). The *Journal of Robotic Surgery* and the *International Journal of Medical Robotics and Computer Assisted Surgery* made notable contributions with 188 and 93 articles, respectively. Additionally, a wide range of journals addressing different aspects of oncology were identified, such as *Diagnostics* and *Journal of Clinical Medicine*. Furthermore, other general-purpose journals like *PLOS One* and *IEEE Access* have also contributed to this developing field (Table 3). In terms of temporal trends, the publications showed variation across

Table 2
Top 30 countries leading AI research in oncology (2003–2023): Publication counts and global share (P%) from 11,149 total publications.

Countries	Articles	P (%)
China	2877	25,81
The USA	2270	20,36
India	733	6,57
South Korea	695	6,23
Japan	550	4,93
Italy	516	4,63
Germany	340	3,05
The United Kingdom	285	2,56
Taiwan	236	2,12
Canada	218	1,96
France	213	1,91
Australia	169	1,52
The Netherlands	163	1,46
Turkey	150	1,35
Iran	139	1,25
Arabia	139	1,25
Spain	133	1,19
Pakistan	127	1,14
Sweden	70	0,63
Brazil	68	0,61
Egypt	60	0,54
Greece	57	0,51
Singapore	52	0,47
Turkey	51	0,46
Romania	50	0,45
Israel	50	0,45
Switzerland	48	0,43
Denmark	48	0,43
Finland	47	0,42
Malaysia	45	0,40

Retrieved from the Clarivate Analytics' Citation Index Expanded database.

the years they were first published, suggesting that the research on AI in oncology is still evolving.

3.2. Latent Dirichlet Allocation (LDA)

From 2003 to 2023, a broad spectrum of AI research applications in oncology was covered by the theme domains identified in Table 4. There were 32 topics with the highest coherence score in the LDA model (Fig. 2). For each latent topic identified, Table 4 displays the 15 most likely terms and their semantically related label. The five topics with the highest number of publications were Tumor Classification through Image Processing (t_10), Predictive Models and Cohort Evaluation (t_18), Clinical Trials and Molecular Therapies (t_31), Robotics in Oncological Surgery (t_3), and Robotic and Laparoscopic Surgery (t_28). These findings point to important areas of oncological research where AI and medicine meet. The following are some possible domains that the various themes allude to (Table 4): Cervical and breast cancer research (t_1, t_6, t_4, t_9, t_15:, t_19, t_20, t_27, and t_29); Machine learning application in tumor diagnosis and classification (t_2: and t_16); Robotics in oncological surgery (t_3 and t_28); Medical imaging and diagnosis (t_5 and t_11); Genetics and molecular biology (t_13 and t_31); Treatment evaluation and prognosis (t_17, t_18, and t_26); and Socio-economic and cultural impact (t_32).

We determined the document-topic distribution, and the average probability of all articles published each year (Eq. (1)) to examine trends over time. Fig. 3 reveals the dynamic landscape of the 32 topics. Both positive and negative regression slopes were statistically significant ($p < 0.01$). Among the topics showing an upward tendency were the tumor classification through image processing (t_10), the development of predictive models and cohort evaluation (t_18), and the focus on quality of life and patient care (t_21). Furthermore, early detection of breast cancer (t_6), patient prognosis (t_9), data splitting for training and testing (t_22), treatment response and clinical data (t_26), pathology and imaging diagnosis (t_29), and colon and rectal cancer detection (t_30)

Table 3
Top 30 journals publishing AI research in oncology (2003–2023): Article counts, total citations, and year of first publication.

Source	Abbreviation	NP	TC	PY_start
Frontiers in Oncology	Front Oncol	467	4210	2012
Cancers	Cancers	399	4047	2019
Scientific Reports	Sci Rep	316	7897	2016
Journal of Robotic Surgery	J Robot Surg	188	1229	2007
Diagnostics	Diagnostics	186	1656	2018
Surgical Endoscopy and other Interventional Techniques	Surg Endosc Other Interv Tech	153	5378	2006
European Radiology	Eur Radiol	128	3095	2015
Ieee Access	IEEE Access	110	1880	2018
Plos One	PLoS One	104	1868	2011
International Journal of Medical Robotics And Computer Assisted Surgery	Int J Med Robot Comput Assist Surg	93	1215	2007
Annals of Surgical Oncology	Ann Surg Oncol	89	3256	2006
Jco Clinical Cancer Informatics	JCO Clin Cancer Inform	89	789	2017
Gynecologic Oncology	Gynecol Oncol	87	4243	2008
Head And Neck-Journal For The Sciences And Specialties Of The Head And Neck	Head Neck	83	1881	2009
Computers in Biology and Medicine	Comput Biol Med	78	2029	2015
Medical Physics	Med Phys	73	1317	2007
Applied Sciences-Basel	Appl Sci	70	698	2018
Journal of Clinical Medicine	J Clin Med	70	609	2019
Multimedia Tools And Applications	Multimed Tools Appl	63	554	2019
Sensors	Sensors	57	1552	2018
Journal of Cancer Research and Clinical Oncology	J Cancer Res Clin Oncol	56	507	2017
International Journal of Molecular Sciences	Int J Mol Sci	55	407	2018
Bmc Cancer	BMC Cancer	54	704	2012
International Journal of Gynecological Cancer	Int J Gynecol Cancer	52	1075	2010
Frontiers In Immunology	Front Immunol	49	270	2021
Journal of Magnetic Resonance Imaging	J Magn Reson Imaging	48	1019	2010
Urology	Urology	48	1043	2008
European Journal Of Radiology	Eur J Radiol	48	968	2018
Frontiers in Genetics	Front Genet	48	475	2019
Journal of Personalized Medicine	J Pers Med	48	361	2020

The table is arranged in descending order according to NP.
Abbreviations: TC, total citations; NP, number of publications; and PY_start, year of publication start.

exhibit promising positive trends. Conversely, several domains exhibited an unfavorable trend, such as research concerning prostate cancer (t_20), robotic and laparoscopic surgery in oncology (t_28), and socio-economic and cultural impact (t_32).

3.3. HJ-biplot

Using multivariate HJ-Biplot analysis, the analysis examined topics over years, journals, and countries, producing a theta matrix of findings. Probability values ranging from 0 to 1 were present in every output matrix. Data variability accounted for 73% in the relationships between topics and years, 65% in the relationships between topics and countries, and 50% in the relationships between topics and journals. The results demonstrated consistent patterns, as illustrated in Fig. 4.

Considering the topics assigned based on research conducted over the years, Fig. 4A shows that the years 2004 to 2006 indicated in blue, are related to topics and t_2 (Machine Learning in Diagnosis), t_7 (Advances in Oncological Research), t_8 (Pancreatic Cancer and Leukemia), t_12 (Machine Learning in Diagnosis), t_13 (Cell Detection and Classification), t_24 (Confidence Intervals and Comparisons), t_31 (Clinical Trials and Molecular Therapies). Subsequently, the following topics are

Table 4
Discovered topics from 11149 articles on AI in oncology (2003–2023): Topic prevalence and top terms.

topic	N	P(%)	top_terms	Label
t_1	187	2.185	review, cervic, trial, includ, cervic_cancer, search, systemat, thyroid, databas, report, literatur, meta, assess, conduct, random	Cancer Research Review
t_2	553	4.222	machine, machine_learn, learn, algorithm, random, accuraci, support, forest, predict, learn_algorithm, random_forest, vector, svm, classifi, perform	Machine Learning in Diagnosis
t_3	859	5.196	patient, complic, robot, oper, time, dai, hospit, postop, compar, underw, outcom, loss, rate, stai, blood	Robotics in Oncological Surgery
t_4	144	2.077	lung, ct, lung_cancer, nodul, pet, ct_imag, tomographi, nsclc, small, comput_tomographi, scan, cancer, comput, pet_ct, tomographi_ct	Lung Cancer Imaging
t_5	425	3.602	imag, sensit, lesion, specif, malign, mri, sensit_specif, diagnost, accuraci, benign, perform, radiologist, evalu, weight, ultrasound	Imaging Diagnosis
t_6	302	3.631	cancer, breast, breast_cancer, earli, detect, diagnosi, techniqu, women, type, death, stage, diseas, diagnos, rate, screen	Breast Cancer Early Detection
t_7	578	4.727	research, applic, recent, challeng, review, provid, medic, data, field, technologi, advanc, develop, improv, futur, current	Advances in Oncological Research
t_8	32	1.317	pancreat, bone, pdac, leukemia, pancreat_cancer, blood, pc, acut, gastrointestin, diagnosi, eu, neoplasm, pediatr, includ, criteria	Pancreatic Cancer and Leukemia
t_9	371	3.449	risk, patient, surviv, factor, stage, prognost, score, index, year, predict, low, prognosi, o, cox, cohort	Patient Prognosis
t_10	1250	6.467	imag, propos, classif, network, tumor, accuraci, cnn, brain, neural, dataset, detect, neural_network, segment, process, convolut	Tumor Classification Image Processing
t_11	429	3.207	segment, plan, dose, volum, evalu, base, manual, net, radiat, target, gener, similar, dice, coeffici, automat	Radiotherapy Treatment Planning
t_12	27	1.488	hcc, liver, decis, clinic, make, decis_make, carcinoma, hepatocellular, hepatocellular_carcinoma, treatment, clinic_decis, patient, carcinoma_hcc, liver_cancer, hcc_patient	Hepatocellular Carcinoma (HCC) and Decision Making
t_13	264	2.473	cell, tissu, sampl, system, normal, accuraci, diagnosi, classif, healthi, classifi, cancer_cell, detect, cancer, combin, cad	Cell Detection and Classification
t_14	219	2.737	featur, radiom, select, extract, combin, base, featur_extract, radiom_featur, classifi, featur_select, perform, textur, imag, radiom_model, phase	Radiomics for Feature Selection
t_15	116	1.935	node, lymph, lymph_node, metastasi, dissect, patient, cancer, ln, metastas, metastat, node_dissect, stage, en, number, node_metastasi	Metastasis and Prognosis
t_16	404	3.32	ai, artifici, intellig, artifici_intellig, detect, system, diagnosi, screen, diagnost,	Artificial Intelligence in Diagnosis

(continued on next page)

Table 4 (*continued*)

topic	N	P(%)	top_terms	Label
			intellig_ai, imag, base, algorithm, assist, tool	
t_17	212	3.359	patient, treatment, month, year, recurr, median, follow, local, rang, treat, diseas, ag, receiv, therapi, post	Treatment and Recurrence
t_18	952	5.689	model, predict, auc, curv, clinic, patient, perform, cohort, develop, area, characterist, base, valid, receiv, predict_model	Predictive Models and Cohort Evaluation
t_19	54	1.926	tumor, grade, tumour, cell, bc, size, carcinoma, renal, rcc, invas, subtyp, low, tumor_size, cell_carcinoma, histolog	Tumor Characteristics
t_20	200	1.92	prostat, prostat_cancer, pca, radic, cancer, prostatectomi, men, function, radic_prostatectomi, biopsi, psa, rarp, posit, gleason, bladder	Prostate Cancer
t_21	276	3.214	data, cancer, skin, care, health, cost, melanoma, effect, identifi, qualiti, oncologi, clinic, skin_cancer, medic, life	Quality of Life and Patient Care
t_22	116	2.811	test, set, train, valid, data, extern, dataset, test_set, perform, intern, extern_valid, cross, develop, data_set, valid_set	Data Splitting for Training and Testing
t_23	57	2.242	outcom, surviv, rate, term, year, oncolog, long, free, diseas, long_term, compar, oncolog_outcom, recurr, free_surviv, colon	Long-term Survival
t_24	35	1.814	ci, interv, ratio, confid, confid_interv, ci_ci, auc_ci, compar, conclus, interv_ci, identifi, patient, assess, cohort, regress	Confidence Intervals and Comparisons
t_25	197	2.593	group, compar, patient, differ, gastric, gastric_cancer, gastrectomi, match, score, time, postop, term, similar, rg, rate	Patient Group Comparisons
t_26	243	2.667	predict, treatment, respons, data, cancer, drug, therapi, clinic, combin, patient, approach, multi, integr, inform, effect	Treatment Response and Clinical Data
t_27	311	2.14	tor, carcinoma, neck, primari, patient, squamou, oral, cell_carcinoma, squamou_cell, transor, treatment, head, surgeri, head_neck, transor_robot	Head and Neck Cancer and Surgery
t_28	755	4.55	robot, surgeri, resect, laparoscop, rectal, approach, surgic, robot_surgeri, cancer, open, minim, invas, assist, rectal_cancer, minim_invas	Robotic and Laparoscopic Surgery
t_29	294	2.486	slide, pathologist, crc, pathologi, colorect, tissu, colorect_cancer, wsi, annot, stain, level, score, histopatholog, digit, biopsi	Pathology and Imaging Diagnosis
t_30	129	3.256	learn, deep, deep_learn, model, imag, base, learn_model, network, learn_base, neural_network, neural, perform, train, resnet, data	Colon and Rectal Cancer Detection
t_31	833	4.278	gene, express, identifi, cancer, cell, relat, immun, biomark, genom, signatur, mutat, subtyp, prognosi, gene_express, pathwai	Clinical Trials and Molecular Therapies
t_32	325	3.023	case, perform, time, procedur, surgic, system, robot, report, feasibl, singl, renal, partial, min, experi, nephrectomi	Socioeconomic and Cultural Impact

Topics with the best coherence score or semantic quality based on the LDA model.
Abbreviations: t, topic; N, number of publications; P(%), topic prevalence.

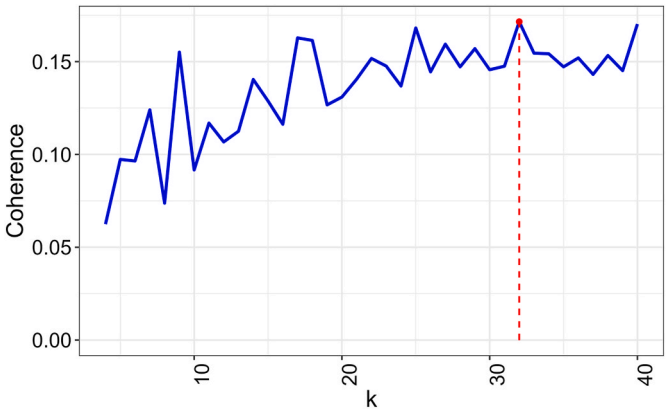


Fig. 2. Evaluation of coherence scores for topic models in oncology across different numbers of topics (k). **Coherence:** Measures the semantic quality of topics. Higher values indicate stronger word relationships; **Topic Models:** Identify thematic patterns in texts; in Oncology, they reveal key research areas; **k:** Represents the number of topics in the model. Helps determine the optimal value for thematic analysis.

grouped in red for the years 2007 through 2017: t_1 (Cancer Research Review), t_3 (Robotics in Oncological Surgery), t_11 (Radiotherapy Treatment Planning), t_15 (Metastasis and Prognosis), t_17 (Treatment and Recurrence), t_20 (Prostate Cancer), t_23 (Long-term Survival), t_25 (Patient Group Comparisons), t_27 (Head and Neck Cancer and Surgery), t_28 (Robotic and Laparoscopic Surgery), and t_32 (Socioeconomic and Cultural Impact). Finally, the years 2018-2023 indicated in green correspond to the following topics: t_5 (Imaging Diagnosis), t_6 (Breast Cancer Early Detection), t_10 (Tumor Classification Image Processing), t_14 (Radiomics for Feature Selection), t_16 (Artificial Intelligence in Diagnosis), t_18 (Predictive Models and Cohort Evaluation), t_19 (Tumor Characteristics), t_21 (Quality of Life and Patient Care), t_22 (Data Splitting for Training and Testing), t_26 (Treatment Response and Clinical Data), t_29 (Pathology and Imaging Diagnosis), and t_30 (Colon and Rectal Cancer Detection).

Iran, Turkey, Pakistan, Egypt, Greece, Malaysia, India, and Saudi Arabia are among the nations grouped in the red in Fig. 4B. These countries are related to topics t_2 (Machine Learning in Diagnosis), t_6 (Breast Cancer Early Detection), t_7 (Advances in Oncological Research), t_8 (Pancreatic Cancer and Leukemia), t_10 (Tumor Classification Image Processing), t_13 (Cell Detection and Classification), t_14 (Radiomics for Feature Selection), t_30 (Colon and Rectal Cancer Detection). Then, under the green group, The United Kingdom, Brazil, Australia, Italy, Spain, the USA, France, Singapore, Taiwan, South Korea, Denmark, Canada, Japan, and Sweden are related to topics t_1 (Cancer Research Review), t_3 (Robotics in Oncological Surgery), t_15 (Metastasis and Prognosis), t_16 (Artificial Intelligence in Diagnosis), t_17 (Treatment and Recurrence), t_20 (Prostate Cancer), t_21 (Quality of Life and Patient Care), t_23 (Long-term Survival), t_24 (Confidence Intervals and Comparisons), t_25 (Patient Group Comparisons), t_27 (Head and Neck Cancer and Surgery), t_28 (Robotic and Laparoscopic Surgery), and t_32 (Socioeconomic and Cultural Impact). Finally, the topics t_4 (Lung Cancer Imaging), t_5 (Imaging Diagnosis), t_9 (Patient Prognosis), t_11 (Radiotherapy Treatment Planning), t_12 (Hepatocellular Carcinoma (HCC) and Decision Making), t_18 (Predictive Models and Cohort Evaluation), t_19 (Tumor Characteristics), t_22 (Data Splitting for Training and Testing), t_26 (Treatment Response and Clinical Data), t_29 (Pathology and Imaging Diagnosis), and t_31 (Clinical Trials and Molecular Therapies) are related to a group of nations, including China, the Netherlands, Switzerland, Finland, Germany, and Israel, represented by blue shade.

The most pertinent journals and the topics they are related to are shown in Fig. 4C. In the blue group, journals like *Frontiers in Immunology*,

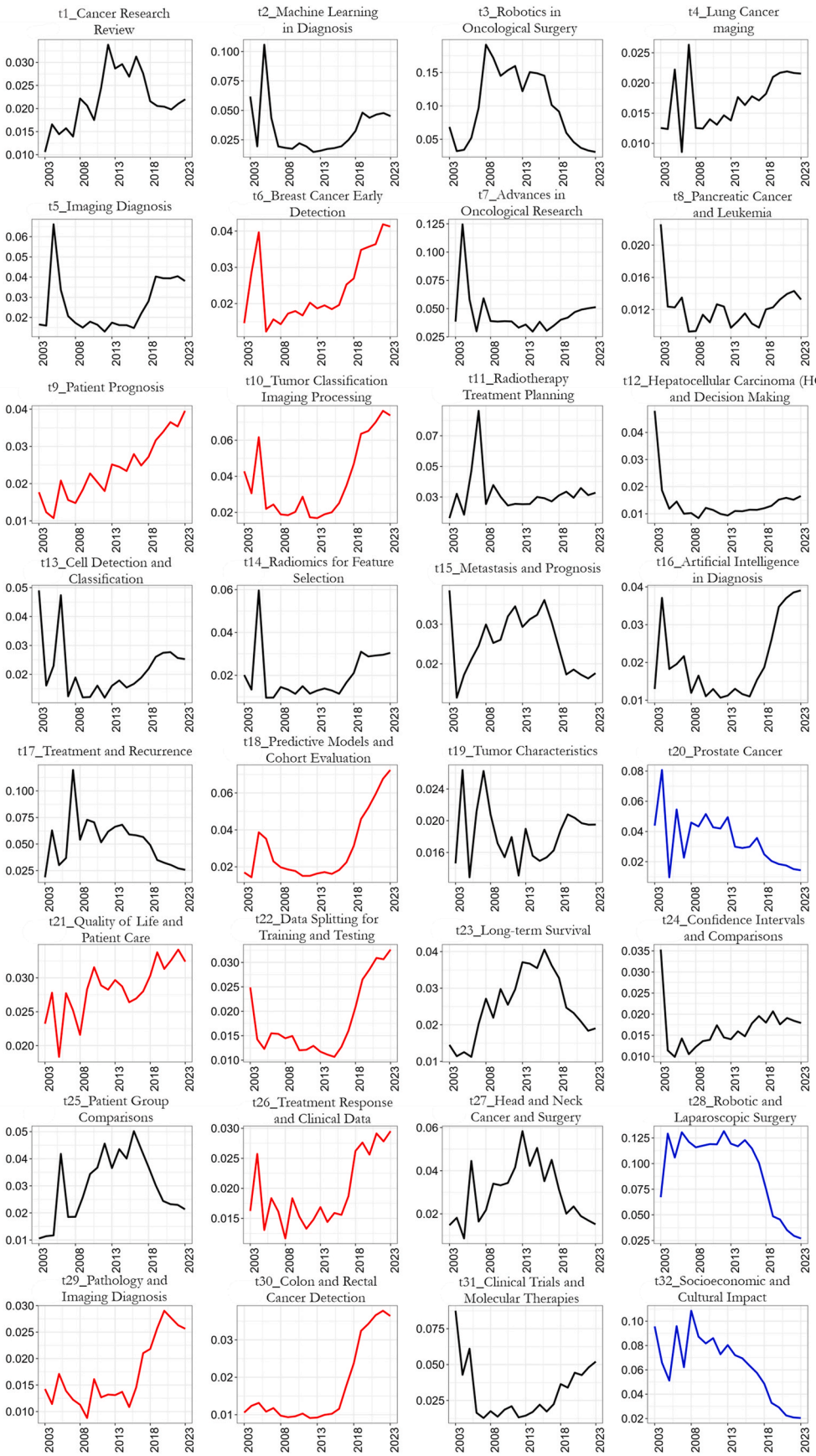


Fig. 3. Trends of research topics in oncology (2003–2023): increasing (red), decreasing (blue), and stable (black) topic dynamics over time.

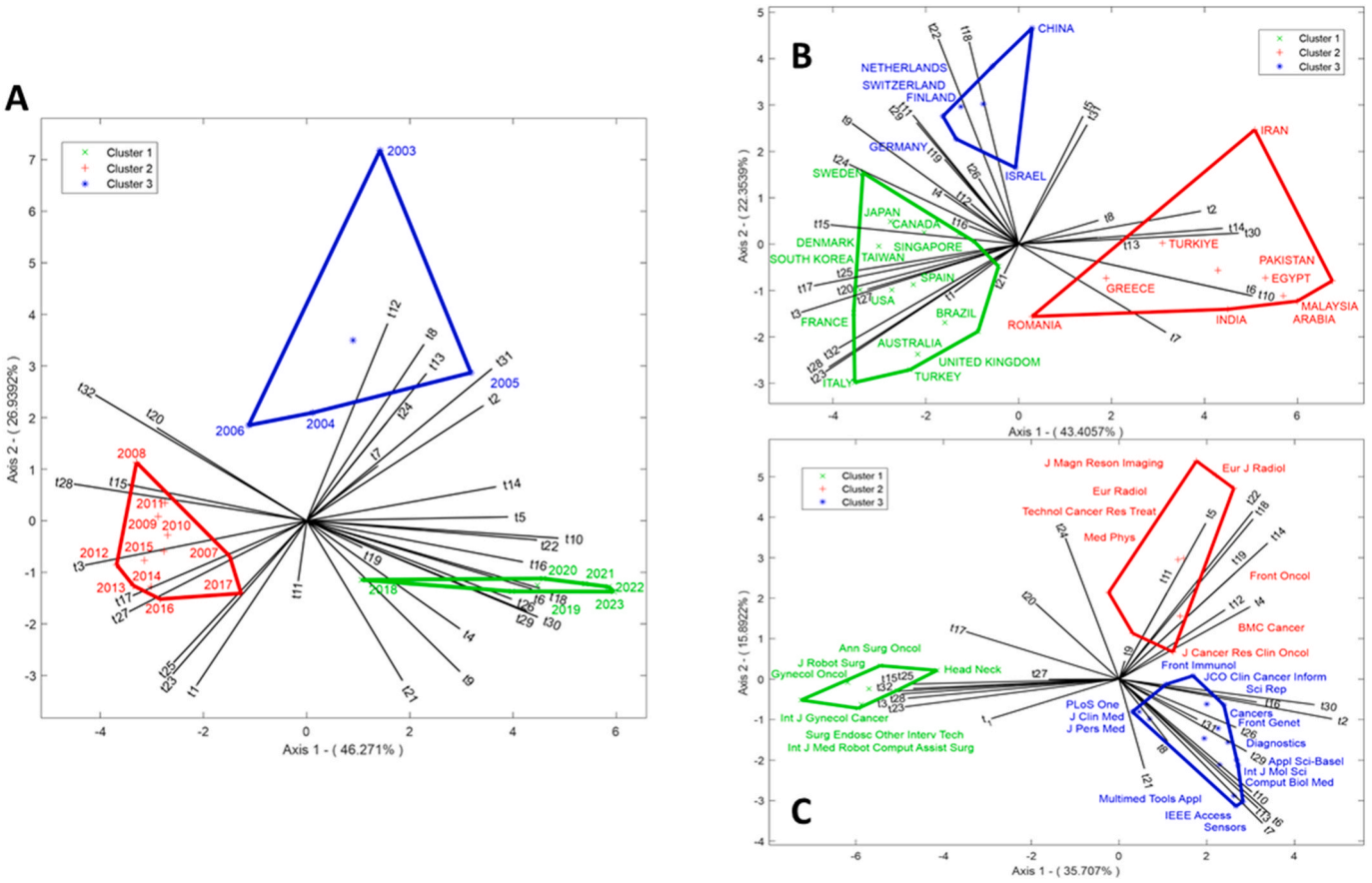


Fig. 4. Comprehensive analysis of topic associations in oncological research using the HJ-Biplot method: Temporal trends by year (a), geographic distribution across countries (b), and contributions by publication sources (c).

JCO Clinical Cancer Informatics, *Scientific Reports*, *Cancers*, *Frontiers in Genetics*, *Diagnostics*, *Applied Sciences (Basel)*, *International Journal of Molecular Sciences*, *Computers in Biology and Medicine*, *Sensors*, *IEEE Access*, *Multimedia Tools and Applications*, *Journal of Personalized Medicine*, *Journal of Clinical Medicine*, and *PLoS One* are mostly associated with topics t_2 (Machine Learning in Diagnosis), t_6 (Breast Cancer Early Detection), t_7 (Advances in Oncological Research), t_8 (Pancreatic Cancer and Leukemia), t_10 (Tumor Classification Image Processing), t_13 (Cell Detection and Classification), t_16 (Artificial Intelligence in Diagnosis), t_21 (Quality of Life and Patient Care), t_26 (Treatment Response and Clinical Data), t_29 (Pathology and Imaging Diagnosis), t_30 (Colon and Rectal Cancer Detection), and t_31 (Clinical Trials and Molecular Therapies). In the green group, journals such as *Annals of Surgical Oncology*, *Journal of Robotic Surgery*, *Head & Neck, Gynecologic Oncology*, *International Journal of Gynecological Cancer*, *Surgical Endoscopy and Other Interventional Techniques*, and *International Journal of Medical Robotics and Computer Assisted Surgery* are related to topics t_1 (Cancer Research Review), t_3 (Robotics in Oncological Surgery), t_15 (Metastasis and Prognosis), t_17 (Treatment and Recurrence), t_20 (Prostate Cancer), t_23 (Long-term Survival), t_25 (Patient Group Comparisons), t_27 (Head and Neck Cancer and Surgery), t_28 (Robotic and Laparoscopic Surgery), and t_32 (Socioeconomic and Cultural Impact). Finally, the group of journals categorized in red, including *Journal of Magnetic Resonance Imaging*, *European Radiology*, *Technology in Cancer Research and Treatment*, *Medical Physics*, *European Journal of Radiology*, *Frontiers in Oncology*, *BMC Cancer*, and *Journal of Cancer Research and Clinical Oncology*, are associated with topics t_4 (Lung Cancer Imaging), t5 (Imaging Diagnosis), t9 (Patient Prognosis), t_11 (Radiotherapy Treatment Planning), t_12 (Hepatocellular Carcinoma (HCC) and Decision Making), t_18 (Predictive Models and Cohort Evaluation), t_19 (Tumor Characteristics), t_22 (Data Splitting for Training and Testing), and t_24 (Confidence Intervals and Comparisons).

4. Discussion

The robust methodology employed in this study, combining machine learning and HJ-Biplot techniques, significantly expanded the scope and depth of the analysis, yielding a higher number of relevant results compared to traditional approaches. The integration of LDA for topic modeling played a pivotal role in uncovering 32 distinct themes with high coherence scores, representing a granular and nuanced analysis of the research landscape. While traditional bibliometric techniques like co-citation and keyword co-occurrence map the surface-level structure of a field—such as research lineage or basic keyword links—they often fail to capture underlying conceptual ideas. Instead of being citation-driven, the LDA algorithm is content (text)-driven. Our approach bypasses these limitations by utilizing LDA to arrange co-occurring words into significant, higher-order themes. Crucially, by pairing LDA with the HJ-Biplot, we moved beyond a simple list of isolated topics. This combined framework allowed us to project these 32 themes simultaneously against metadata, revealing hidden micro-trends and mathematically isolating critical research gaps that citation patterns or keyword frequency counts alone inherently obscure.

Although Latent Dirichlet Allocation (LDA) offers a potent unsupervised method for mapping massive text corpora, it is intrinsically devoid of a native understanding of syntax, local semantic subtleties, and sequential context due to its underlying "bag-of-words" assumption. However, by incorporating the HJ-Biplot statistical method, this risk was specifically reduced in this study. Because the HJ-Biplot maps total structural variance rather than volume, and models the correlations

between terms and topics, it successfully isolated emerging fields and completely under-researched areas. The HJ-Biplot statistical technique further complemented LDA by visualizing relationships between years, countries, journals, and topics in a shared low-dimensional space, offering insights into temporal, geographic, and disciplinary trends.

The HJ-Biplot broke the chronological progression of the field into three fundamental eras (Fig. 4A). First, the nascent era (2004–2006, blue cluster), where researchers were testing early, was characterized by classical machine learning algorithms on limited clinical trial datasets to see if computational models could assist in basic classification tasks. During the following decade (2007–2017, red cluster) AI matured into precision applications in robotic surgery and automation in radiation dosage planning. Powered by the deep learning boom and an abundance of digital imaging data, the current landscape (2018-2023) of AI in oncology focuses on extraction of sub-visual features of medical imaging and highly accurate automated diagnostic pipeline validation. The spatial distribution of countries across the biplot reveals a clear macroeconomic and infrastructure-driven division of labor in global oncology research (Fig. 4B). The red cluster comprises developing and emerging research hubs such as India, Malaysia, and Saudi Arabia levering open-source AI frameworks to address massive public health pressures, focusing on scalable, cost-effective screening and early diagnostic classification tools. Western Europe, North America, Australia, and advanced Asian economies (e.g., South Korea, and Japan) possess the heavy capital infrastructure required to pioneer expensive surgery, alongside mature healthcare systems focusing on post-treatment survival metrics and socioeconomic impacts. The blue cluster led by China is the vanguard of data-driven oncology and precision medicine. The journal-topic biplot maped the academic channels driving the dissemination of oncology AI, dividing the journal focus areas into three clear editorial lines (Fig. 4C). The first is the computational and open-access generalist hub (blue cluster) that prioritizes new algorithms, neural network architectures, and technical proofs-of-concept. The second is the clinical-specific hub (green cluster) focusing on the practical utility accuracy, and safety of AI tools in Oncology. Finally, the targeted imaging and radiotherapy hub (red cluster) focusing on quantitative imaging biomarkers and translating pixel-level computational data into radiotherapeutic diagnosis. The HJ-Biplot method thus offered a strategic roadmap for oncology research.

Additionally, the use of advanced bibliometric tools like Bibliometrix and Textminer ensured methodological rigor, reproducibility, and accuracy. These tools facilitated efficient data cleaning, preprocessing, and the extraction of actionable insights from over 11,000 articles. Together, this integrated approach allowed the study to achieve a significantly higher number of results compared to traditional bibliometric analyses, which often rely solely on citation-based metrics and are limited in thematic exploration. Nevertheless, the study has several limitations, including the subjectivity inherent in manually labeling LDA topics, the potential omission of relevant publications due to reliance a single database, linguistic bias from excluding non-English articles, the limitations of the "bag-of-words" model that ignores grammar and context, and the exclusion of grey literature such as book chapters. The Web of Science (WoS) Core Collection database is recognized as the gold standard for metadata and textual formatting consistency, providing exceptionally rigorous representation of the authoritative core text of the AI-oncology environment. To improve the generalizability of our findings, future studies should include a wider range of databases and incorporate non-English publications to minimize bias. Lastly, although a rigorous six-expert consensus protocol reduced human subjectivity in topic labeling, future technical iterations of this hybrid framework could investigate fully automated subject-naming methods using cutting-edge NLP models, such as transformer-based embedding pipelines (e.g., BERTopic) or LLM-driven zero-shot labeling.

This exploratory research identified some research topics where AI and oncology intersect by year, source, and country. The most important research areas that support accurate diagnosis, treatment, recurrence, and survival predictions in oncology were the "Tumor Classification-Image Processing" (t_10), "Predictive Models and Cohort Evaluation" (t_18); "Treatment Response and Clinical Data" (t-26), "Pathology and Imaging Diagnosis" (t_29), and "Clinical trials and Molecular Therapies" (t_31). Unlike traditional bibliometric analysis, our machine learning- and statistical approaches [28,29] dealt with mapping the literature on AI applications in oncology research to unpack the emerging areas of great research interest in that field.

AI uses mathematical-based algorithmic principles like those of the human mind to gain deep knowledge underlying big datasets with great potential in health care [37]. AI probes outperform conventional statistical techniques in aiding in the early detection and prognosis of cancer, a malignancy with numerous etiologies [38,39]. The first prominent area of research focus, the precise tumor categorization via image analysis is essential (t_10 and t_29) because it can significantly impact critical clinical choices, including treatment options and patient outcomes. Automated image interpretation and computer-assisted diagnosis of tumors continue to be a topic of burgeoning field [40] with enormous potential in guiding precision cancer therapy [41,42]. The optimization of the clinical-radiological workflow (patient screening, image capture), as well as more specific image-based tasks (cancer identification, characterization, and treatment monitoring), are all part of the integration of AI into tumor imaging. Breast cancer is undoubtedly a major field of AI development in radiological screening practices (t_6) [43,44]. Using DL algorithm as a second reader enhances radiologists' performance in the detection of lung cancer [45]. Deep learning techniques like convolutional neural networks (CNN) have successfully been used for segmentation tasks, one of the most difficult aspects of oncological image analysis, to obtain precise outcomes for planning, radiotherapy, measuring volumes, and tracking the evolution of cancer [46,47]. AI is being used to investigate the temporal follow-up of malignancies and treatment responses to develop precise models for the assessment of effective anticancer therapies [48,49]. As future challenges, the lack of a consistent imaging process [46] and the need for reliable and robust validation techniques [50] remain barriers to the clinical application of AI-assisted cancer imaging analysis.

Because AI methodologies can analyze a large volume of different types of data, such as clinical risk factors, genetic data, and imaging examinations [46,51], they can help with cohort evaluation (t_18). AI makes it possible to combine clinical and imaging data to improve the effectiveness of screening programs and patient prognosis (t_9), which will, without a doubt, bridge the gap between model development and clinical implementation [52,53]. Similarly, because of the increasing focus on understanding the physiopathology of cancer and developing more specialized and effective treatments [54], the topic of treatment response and clinical data is also highly important (t_26). By combining biochemical and clinical data, AI-powered models have the potential to stratify patient risk and guide personalized treatment plans, resulting in the best possible onco-functional balance [55–57]. Finally, clinical trials and molecular therapies (t_31) provide evidence that the cornerstone of precision medicine for cancer treatment has been molecular targeted therapy, which uses monoclonal antibodies and small-molecule medicines that operate as signal transduction inhibitors [58]. These methods are currently applied in clinical settings as initial treatments for a variety of human malignancies [59].

A pivotal role of robotics in oncological and laparoscopic surgery (t_3) was also identified by our analysis as another important topic in the AI-oncology interplay. This technological advancement improves the accuracy and safety of surgical operations, which may have a significant impact on postoperative results and patient recovery [60]. Surprisingly, the field of robotic and laparoscopic surgery (t_28) demonstrated a decline in oncology, despite the growing interest in colorectal [61] and endometrial [62] cancers. To meet opportunities and challenges in the field of oncology, unfavorable trends (Fig. 3) highlighted additional areas —prostate cancer and socioeconomic and cultural impacts —that may need more attention and research investment in AI [63,64].

By identifying hidden features, AI software can completely transform oncology predictions in comparison to previously published approaches. Applications of AI in oncology include, but are not limited to, deepening the understanding of tumor biology, enhancing clinical practice and care delivery (prognosis, diagnosis, and precision oncology), and optimizing cancer research [4–6]. Examining innovative clinical pilot programs where these AI models have successfully transitioned into active oncological care is informative to comprehend the practical implications of the 32 trends revealed by our methodology. Large-scale prospective clinical trials in the field of automated imaging, like the Mammography Screening with Artificial Intelligence (MASAI) trial, have shown that using AI as an autonomous second reader in radiology workflows can safely reduce workload by more than 40% without sacrificing diagnostic sensitivity [65]. Similarly, the use of AI-driven curation platforms in multidisciplinary Molecular Tumor Boards in Precision Oncology has moved from theory to reality; these tools quickly synthesize multi-modal genomic and clinical data to suggest customized molecular therapies for refractory cancer cohorts [66]. Lastly, the delineation of healthy organs at risk has been successfully automated in Radiation Oncology using clinical integrations of deep-learning segmentation models, reducing a multi-hour manual planning process to minutes [67].

The translation of oncological AI applications into routine clinical practice still faces significant ethical, technical, and logistical hurdles. A major bottleneck is data imbalance, when models are trained on restricted cohorts, algorithmic bias is introduced, reducing the generalizability of predictive models. Another issue is clinical validation, since many studies rely on retrospective datasets, AI tools must undergo rigorous, prospective, multi-center clinical trials to safely guide precision oncology. Integrating electronic health and medical records requires robust data-sharing frameworks. Although our analysis shows that computer-assisted tumor classification is a leading emerging field, the lack of standardization in medical imaging fundamentally limits its systemic scalability. Algorithmic brittleness, in which deep learning models experience significant performance drops when exposed to data from different scanner manufacturers or different institutional protocols, necessitates the development of rigorous, vendor-neutral preprocessing pipelines. Moreover, widespread adoption necessitates the maturation of advanced verification techniques, such as robust uncertainty estimation and explainable AI architectures, to confirm that semantic classifications are based on actual pathological features rather than confounding digital artifacts. Ensuring patient confidentiality, compliance with global data protection regulations, and protecting against cyber threats are paramount ethical duties that the current literature often underreports. Crucially, our study explicitly identified that AI cannot truly revolutionize oncology if it ignores the sociocultural architecture of healthcare delivery. Future research must not only develop AI algorithms but also investigate implementation frameworks that ensure equitable deployment.

5. Conclusions

In contrast to typical systematic reviews and bibliometric analyses, our AI-based approach for bibliographic research allowed documents to communicate the breadth and flexibility of AI applied to cancer based on their statistical properties rather than human reviewers, adding new insights to the corpus of existing research. Even though the field of AI-assisted medical imaging analysis, particularly radiology and pathology, has made great strides, our analysis revealed that its integration into lab techniques in the fields of clinical multi-omics, biochemistry, and hematology promised additional oncology improvements and precision oncology. The use of robotics in oncological and laparoscopic surgery is another fascinating new development in intelligent oncology that has unexpectedly been overlooked. Lastly, our investigation also identified knowledge gaps such as the continued lack of AI analysis on the relationship between socioeconomic and cultural factors and cancer prevalence. In summary, our findings imply that more comprehensive and interdisciplinary patterns will be beneficial for the clinical implementation of AI devices in real-world oncology.

Statement on the use of generative AI

The authors attest that this work was prepared without the assistance of generative artificial intelligence (AI) or AI-assisted technologies.

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CRediT authorship contribution statement

Javier De La Hoz-M: Conceptualization, Investigation, Methodology, Writing – original draft. **Karime Montes-Escobar:** Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft. **Carlos Alfredo Salas-Macias:** Conceptualization, Software, Validation, Writing – original draft. **Martha Fors:** Formal analysis, Validation, Writing – review & editing. **Santiago Ballaz:** Formal analysis, Investigation, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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